AEO Citation Pipeline — Design Document

Milestone 2: Problem Formalization and System Design

Natalia Kusmirek | Readings in Computer Science | Instructor: Prof. Maíra Marques Samary

1. Cohort Taxonomy

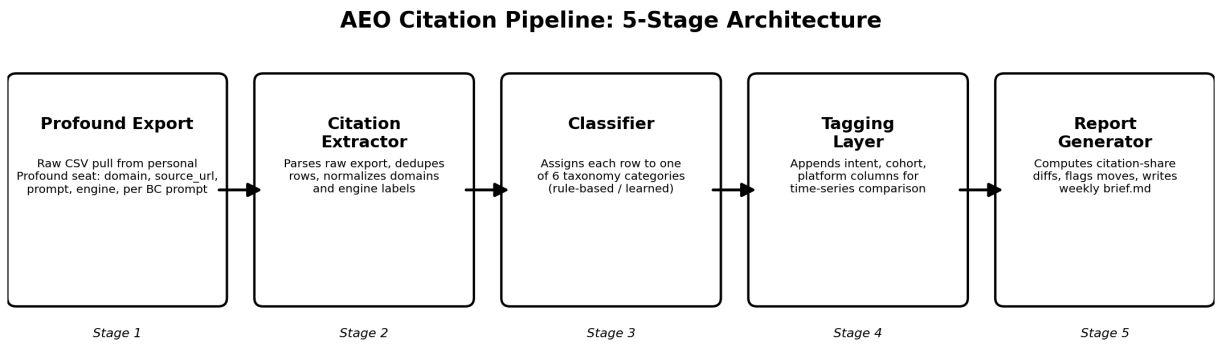
The classifier developed in Milestone 3 sorts every domain that cites Boston College across AI answer engines into one of six mutually exclusive categories. The table below defines each category and the rule used to assign it.

Category	Rule
BC-owned	The domain is a Boston College-controlled property (bc.edu and subdomains, official BC social accounts, BC-run publications).
Earned media	The domain is a third-party news, magazine, or broadcast outlet that published independently about BC without BC controlling the content.
Education directory / ranking site	The domain's primary function is aggregating or ranking schools/programs (e.g., Niche, Peterson's, College Navigator, US News rankings pages).
Forum / community	The domain hosts user-generated discussion (Reddit, College Confidential, Quora, and similar indexed community content).
Competitor institution	The domain belongs to another college or university cited alongside or instead of BC (e.g., bu.edu, nd.edu, holycross.edu).
Other	The domain does not fit the above five categories (e.g., .gov sites, Wikipedia, employer/career sites).

Edge case note: US News is split by page type — its editorial/news articles about BC are labeled earned media, while its rankings pages are labeled education directory/ranking site. This split is applied consistently during manual labeling in Section 3.

2. Pipeline Architecture

The full system is a 5-stage pipeline. Stages 1-2 (Profound export, citation extractor) are built and populated in this milestone; the classifier (Stage 3) is built in Milestone 3; the tagging layer and report generator (Stages 4-5) are built in Milestone 4.



Stage 1 - Profound Export: the 26 prompts in Section 3 are run through a personal Profound seat; citations are exported as a raw CSV with columns domain, source_url, prompt, engine.
Stage 2 - Citation Extractor: parses the raw export, deduplicates rows, and normalizes domain and engine formatting.
Stage 3 - Classifier: assigns each row one of the six taxonomy categories (rule-based baseline and a learned model, built in Milestone 3).
Stage 4 - Tagging Layer: appends intent, cohort, and platform columns so weekly snapshots remain statistically comparable over time.
Stage 5 - Report Generator: computes citation-share deltas between snapshots, flags material moves, and writes a formatted weekly brief.

3. Full Prompt List (26 prompts)

Prompts are split across three intent types to ensure the resulting citation set reflects how real users query AI answer engines about Boston College. Informational and comparison prompts name Boston College directly, since these intents presuppose the user already knows to ask about BC specifically. Recommendation prompts deliberately do not name BC; they are category-level queries (Jesuit/Catholic education, rigorous academics, holistic education, undergraduate education) sourced from Profound's auto-generated prompt set for BC's brand profile. This tests whether BC is cited organically when a user has not already named it, which is the more realistic test of recommendation-intent citation behavior and the stronger methodological choice for this category. Each prompt is also tagged with a Topic and one or two target Personas (Prospective Student, and Parent of Prospective Student where the topic is one a parent would plausibly search), matching the metadata loaded into the Profound prompt set.

Informational (10)

Topic	Prompt	Personas
BC academic reputation	What is Boston College known for academically?	Student
BC admissions	What is the acceptance rate at Boston College?	Student + Parent
BC computer science program	What majors does Boston College offer in computer science?	Student
BC campus life	What is campus life like at Boston College?	Student + Parent
BC cost of attendance	How much does it cost to attend Boston College?	Student + Parent
BC history	What is the history of Boston College?	Student
BC financial aid	What financial aid does Boston College offer?	Student + Parent

Topic	Prompt	Personas
BC business school	What is the Carroll School of Management known for?	Student
BC student housing	What housing options are available for BC freshmen?	Student + Parent
BC study abroad	What study abroad programs does Boston College have?	Student

Comparison (8)

Topic	Prompt	Personas
BC vs Boston University	How does Boston College compare to Boston University?	Student + Parent
BC vs Northeastern	Is Boston College better than Northeastern for business?	Student
BC vs Notre Dame	Boston College vs. Notre Dame: which has a stronger Jesuit tradition?	Student + Parent
BC vs UMass Amherst	How does BC's computer science program compare to UMass Amherst's?	Student
BC vs Holy Cross	Boston College vs. Holy Cross: which is a better fit for a pre-law student?	Student
BC cost vs Catholic universities	How does the cost of Boston College compare to other private Catholic universities?	Student + Parent
BC vs Villanova	BC vs. Villanova: which has better career outcomes in finance?	Student
BC vs Boston University	How does Boston College's campus compare to Boston University's in terms of location?	Student + Parent

Recommendation (8)

Topic	Prompt	Personas
Jesuit/Catholic higher education tradition	What are the best Jesuit colleges for combining academics with community service?	Student
Jesuit/Catholic higher education tradition	Where can I find a Catholic college with an active faith community?	Student + Parent
Jesuit/Catholic higher education tradition	Who offers the strongest financial aid for a Catholic college education?	Student + Parent
Rigorous academics	What are the best colleges for a challenging liberal arts education?	Student
Rigorous academics	Who offers the best undergraduate business education with challenging coursework?	Student
Holistic education (intellectual, ethical, and spiritual development)	Where can I find a college centered on service to others?	Student + Parent
Holistic education (intellectual, ethical, and spiritual development)	What are the best colleges for discussing life's big questions?	Student
Undergraduate education	What are the best colleges for studying business as an undergraduate?	Student